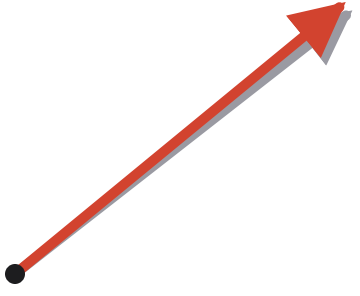
1 **The probe changes the user token's hidden state by 2.5%**



2 **Training shrinks head 6's key for the user token to the probe's scale**

key length, original model

70

key length, trained

6.6

change made by the probe

4.2

3 **Keys are rescaled to full length: only the probed key outscores the rest**

score for user, no probe

4.9

score for user, probe

16.8

all other tokens
combined: 10.9

4 **Head 6 reads user and the answer becomes Yes**

No probe

`</think>` → **No**

0.3% on user

Probe attached

`user` → **Yes**

99.7% on user